

Exercises:

Task1:

Stroke Prediction Dataset: <https://www.kaggle.com/datasets/fedesoriano/stroke-prediction-dataset>

Q1. Basic Data Inspection [4 marks]

- a. Load the dataset using `pandas.read_csv()`
- c. Use `df.info()` and `df.describe()` to summarize the dataset
- d. Show the number of missing values in each column

Q2. Handling Missing Data [4 marks]

- a. Drop any column with **more than 30% missing values** (if any)
- b. For remaining missing values:
 - Fill **numerical** columns (like `bmi`) with the **median**
 - Fill **categorical** columns (if any are missing) with the **mode**
- c. Confirm that no missing values remain using `df.isnull().sum()`

Q3. Handling Duplicates [2 marks]

- a. Check if the dataset contains any **duplicate rows**
- b. Remove duplicates using `drop_duplicates()` and report how many were removed

Q4. Encoding Categorical Variables [4 marks]

- a. List all the **categorical columns**
- b. Apply **One-Hot Encoding** to all **nominal** categorical columns
- c. Apply **Label Encoding** to any **ordinal** categorical column (if applicable)

Q5. Outlier Detection & Removal [5 marks]

- a. Detect **outliers** in `bmi` and `avg_glucose_level` using the **IQR method**
- b. Detect outliers in the same columns using the **Z-score method**

- c. Report how many rows were removed by each method
- d. Compare both techniques and explain which is more suitable for this dataset and why

Task2: (11)

Visual Exploration of the Iris Dataset

The Iris dataset contains measurements of three species of iris flowers: *setosa*, *versicolor*, and *virginica*. Your goal is to visually explore the dataset to understand relationships and distributions among features.

- **Load the dataset** using seaborn or sklearn.datasets.
- **Display basic information:** shape, column names, and the first five rows.
- **Plot a scatter plot** to show the relationship between:

sepal_length and sepal_width, color-coded by species.
- **Plot box plots** for each numerical column (sepal_length, sepal_width, petal_length, petal_width) to visualize spread and detect outliers, separated by species.
- **Plot histograms** for each numeric feature to explore their distributions.
- **Plot a correlation matrix (heatmap)** to understand how the features are correlated.